Poll Everywhere Quizzes

At the beginning of each lecture:

1. Use your device to log in to PollEverywhere (PE). Enter your ANU email (uXXXXXXX@anu.edu.au); PE supports ANU SSO.

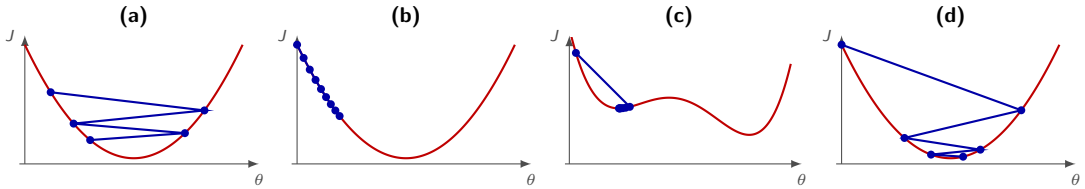
   https://pollev.com/anuiml2026



Photographing or recording the quiz questions is not permitted

Lecture 12 - Quiz 1 (120 seconds)

L12Q1 - Each panel shows the iterates of gradient descent $\theta \leftarrow \theta - \alpha J'(\theta)$ on an error $J(\theta)$. Which statement is correct?

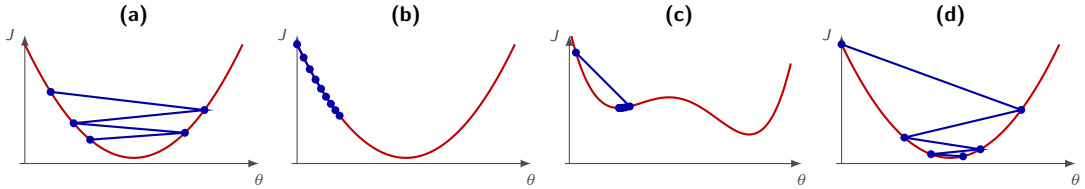


- A) (d) has the same problem as (a): the iterates cross the minimum, so α is too large and must be reduced
- B) (c) has the same problem as (b): α is too small, so the iterates stopped before reaching the lowest valley
- C) (b) will never reach the minimum: with a fixed α, gradient descent only gets within a distance proportional to α
- D) (d) needs no fix, and in (c) gradient descent converged, to a local minimum, because J is not convex

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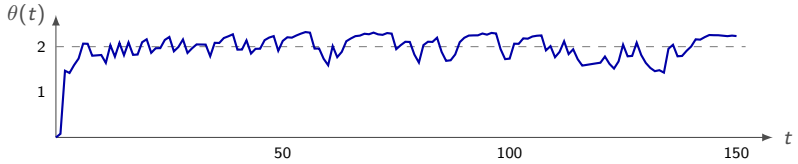


- A) (d) has the same problem as (a): the iterates cross the minimum, so α is too large and must be reduced
- B) (c) has the same problem as (b): α is too small, so the iterates stopped before reaching the lowest valley
- C) (b) will never reach the minimum: with a fixed α, gradient descent only gets within a distance proportional to α
- **D) (d) needs no fix, and in (c) gradient descent converged, to a local minimum, because J is not convex** *In (d) each step multiplies the distance to the minimiser by a factor between -1 and 0, so the iterates converge. In (c) they stop where $J' = 0$, and a smaller α stops at the same point. In (b) they converge, only slowly.*

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Lecture 12 - Quiz 2 (120 seconds)

L12Q2 - SGD with mini-batches of one observation and a constant learning rate α is run on the one-parameter running example. The dashed line is the minimiser $\hat{\theta} = 2$. Which change would fix the problem shown in the plot?

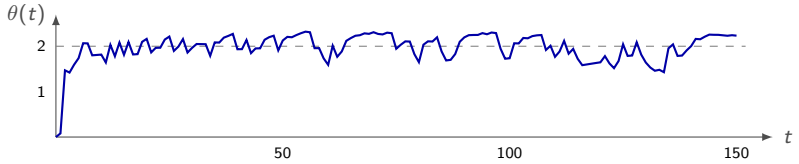


- A) Run more iterations: each SGD step is right on average, so the fluctuations cancel out
- B) Use a smaller constant learning rate from the start: the iterates then settle at $\hat{\theta}$, only later
- C) Keep α large at first and decrease it as t grows: the iterates arrive quickly, then settle
- D) Use mini-batches of two observations: the average of two gradients is exact, so the iterates settle

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Lecture 12 - Quiz 2 (120 seconds)

L12Q2 - SGD with mini-batches of one observation and a constant learning rate α is run on the one-parameter running example. The dashed line is the minimiser $\hat{\theta} = 2$. Which change would fix the problem shown in the plot?



- A) Run more iterations: each SGD step is right on average, so the fluctuations cancel out
- B) Use a smaller constant learning rate from the start: the iterates then settle at $\hat{\theta}$, only later
- **C) Keep α large at first and decrease it as t grows: the iterates arrive quickly, then settle** *With a constant α, SGD wanders around $\hat{\theta}$ however long it runs. A smaller constant α still wanders, only less, and arrives later. A mini-batch of two reduces the noise but does not remove it.*
- D) Use mini-batches of two observations: the average of two gradients is exact, so the iterates settle

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